

DUY NGUYEN

D A T A E N G I N E E R

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PROFILE

As a seasoned Senior Data Engineer with a robust background in Software Engineering, I possess a wealth of expertise in Automation, Backend Development, Artificial Intelligence, and Data Processing & Visualization. I am passionate about leveraging my skills to solve complex problems and drive innovation in the field of data.

SKILLS

- Python, Go, SQL, ReactJS
- Data Warehouse, LakeHouse
- Extract, Transform, Load (ETL)
- Document Processing
- Airflow, Spark, Kafka, Superset
- CI/CD
- Kubernetes, Docker
- AWS, GCP
- Problem-Solving
- Strong Communication

EXPERIENCE

SENIOR DATA ENGINEER

Vietnam Silicon (Apr 2025 - Present) - <https://vnsilicon.net/>

- Collaborate with senior engineers to design and build modern data platforms on AWS and Open-Source Software (OSS), supporting both streaming and batch processing using Airflow, Glue, EMR, Redpanda, EKS, etc.
- Drive the implementation of a modern data platform for the Durian Traceability (DT) project, enabling the development of lakehouse analytics and advanced ML/DL use cases such as fraud detection, voice recognition, and Thailand OCR processing.
- Project Leader for the company's Internship Program Batch 1 — responsible for mentoring, career guidance, technical training, talent evaluation, and recruitment support.

EDUCATION

BACHELOR'S DEGREE OF ELECTRICAL (TELECOMMUNICATION)

Ho Chi Minh city University of technology

Programme de Formation d'Ingénieurs d'Excellence au Vietnam (2019)

SENIOR DATA ENGINEER

Renova Cloud (Sep 2024 - Apr 2025) - <https://renovacloud.com/>

- Provide consulting and solutions for building scalable data platforms on AWS, leveraging native services such as Redshift, S3, Glue, EMR, EKS, and MSK.
- Provide consultation solutions for migrating, relocating, and refactoring on-premises data platforms (HDFS, Spark, Superset, etc.) to AWS, ensuring minimal downtime and cost-efficiency.
- Work directly with clients to manage expectations and ensure project timelines are met, aligning solutions with business needs.

LEAD DATA ENGINEER - PARTIME

MyLong (Jun 2024 - Oct 2025) - <https://www.opusmatch.ai/>

- Design, build, and maintain a Data Platform on AWS using services like Redshift, S3, Glue, EMR, Airflow, and DMS to optimize workflows and meet business requirements.
- Set up near real-time data processing systems with AWS services (Firehose, DMS, Aurora) to handle high-volume requests, supporting recommendation systems in healthcare.
- Collaborate with Data Scientists to enhance AI models and analytics for healthcare applications.
- Work closely with DevOps engineers to optimize deployment pipelines, automate infrastructure, enhance system observability, and improve platform reliability and performance.
- Lead a Data Engineering team, ensuring alignment with project goals and driving innovation.

SENIOR DATA ENGINEER

VieON (Sep 2023 - Sep 2024) - <https://vieon.vn/>

- Lead migration of current Data Platform (HDFS, Spark, Superset, etc.) to Kubernetes. Ensure minimal downtime and optimal performance with regular maintenance. Establish monitoring and alert systems.
- Lead 4-person Data Engineer team, providing guidance and fostering innovation. Align team activities with project objectives.
- Architect's scalable platform (Trino, MinIO, Kafka, Spark, Spark Streaming, Superset, etc.) for 5+ companies, handling 2+ TB data per day. Collaborate with stakeholders for tailored solutions.
- Maintain and enhance features for Recommendation API. Collaborate with data scientists for scalable solutions.

SENIOR DATA ENGINEER

DIGI-TEXX (Mar 2022 - Sep 2023) - <https://digi-texx.com/>

- Design and build a scalable data pipeline that can handle 1TB images per day, including a data warehouse and data lake based on S3, Hbase, Hadoop, Hive, Superset, and Airflow.
- Design and Implement AI Product on Big Data Infrastructure, including but not limited to: ML, DL, HDFS, Spark, Kafka.
- Implement microservices in AI systems with scalable, efficient code to reduce 70% operational costs.
- Implement Image Classification, Image Denoising, Field Detection and Optical Character Recognition models using AI techniques.
- Managing AI engineering research team.

DATA ENGINEER

DIGI-TEXX (Sep 2020 - Mar 2022) - <https://digi-texx.com/>

- Developed an ETL/ELT system collect more than 100 data sources to replace an existing data pipeline.
- Collaborated with data scientists and product engineers to develop and deploy solutions to customer problems.
- Analytic Data Modeling (Star Schema, SCD's, and Lookups)
- Increased the efficiency of the data fetching by approximately 30% using query optimization, indexing and partition.

BACKEND DEVELOPER / AI ENGINEER

Amaris Consulting (Dec 2019 - Sep 2020) - <https://amaris.com/>

- Proposing solutions and technical design of social ios backend applications.
- Deploy REST-FULL API endpoints with FastAPI on AWS services.
- Build a product recommendation system based on product description with Deep Learning.
- Build bots to automatically collect data from mutiple source.
- Performing analysis to assess quality and meaning of data.

AUTOMATION TESTER

TMA Solution (Apr 2019 - Oct 2019) - <https://www.tmasolutions.vn/>

- Deploy automation framework with selenium on cross-platform android, ios, window, macos and web.
- Research and propose solutions to optimize automation framework.
- Executing telecom system test cases on multiple platforms through automation framework.
- detect bugs, report them to developers and help them reproduce the bug.

MACHINE LEARNING ENGINEER JUNIOR

Data Science Lab (Aug 2018 - Feb 2019)

- Provide technical solutions to detect accident victims in inaccessible terrains such as cracks, large fires, etc. through image analysis
- Development of victim detection model with Deep Learning based on Tensorflow and Python
- Preprocessing data and training model.
- Build a real-time victim detection system with Google Firebase.